

UNIVERSITY OF THIES



UFR-Sciences of Engineer

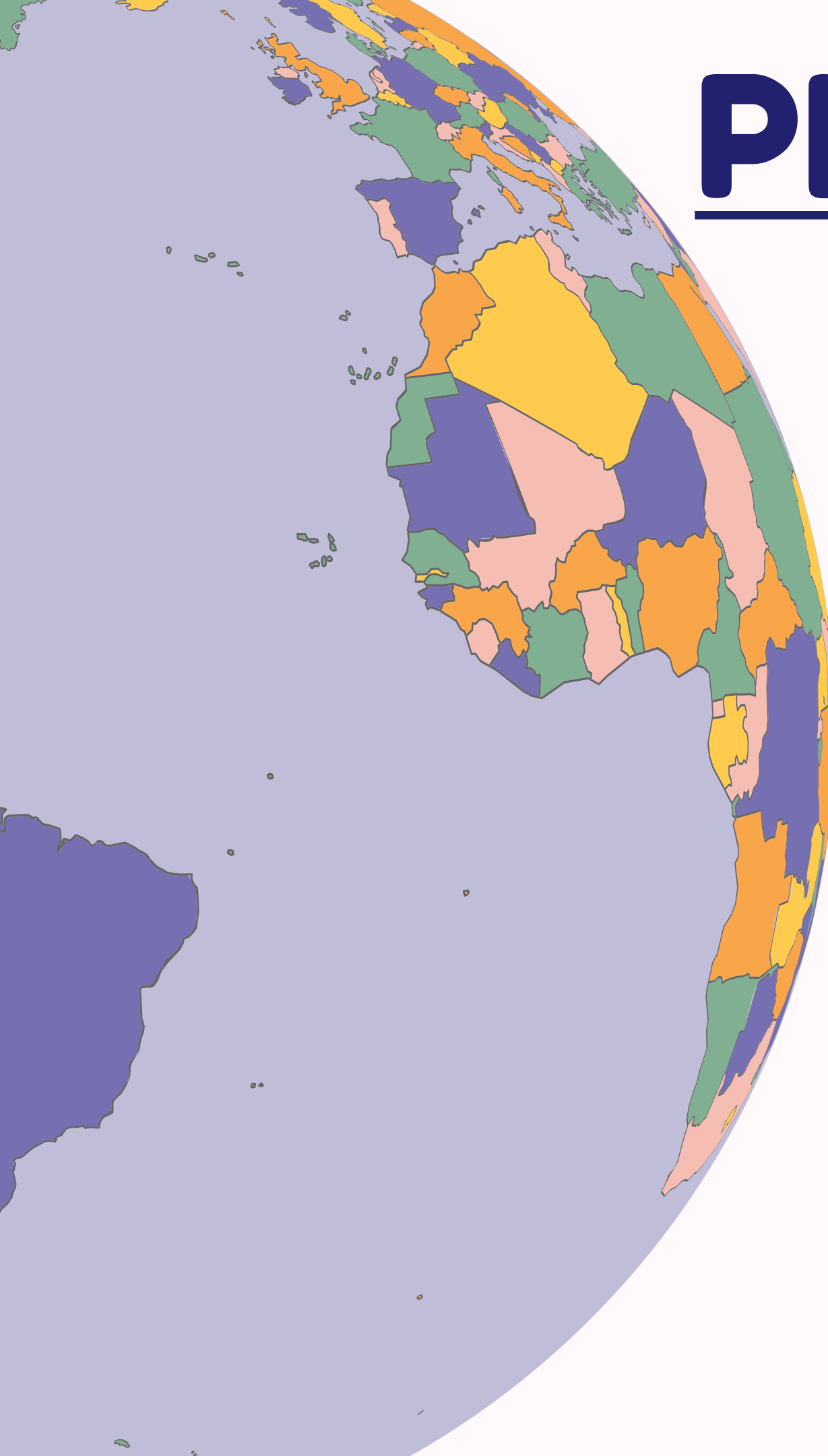
THEME : Cartography and Mapmaking

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PLANNING

Introduction

1. History and Definition of cartography
2. What is Mapmaking ?
3. Tools ,Technologies and Users
4. Importance of Cartography and Mapmaking
5. Advantages and Goals

Conclusion



Introduction

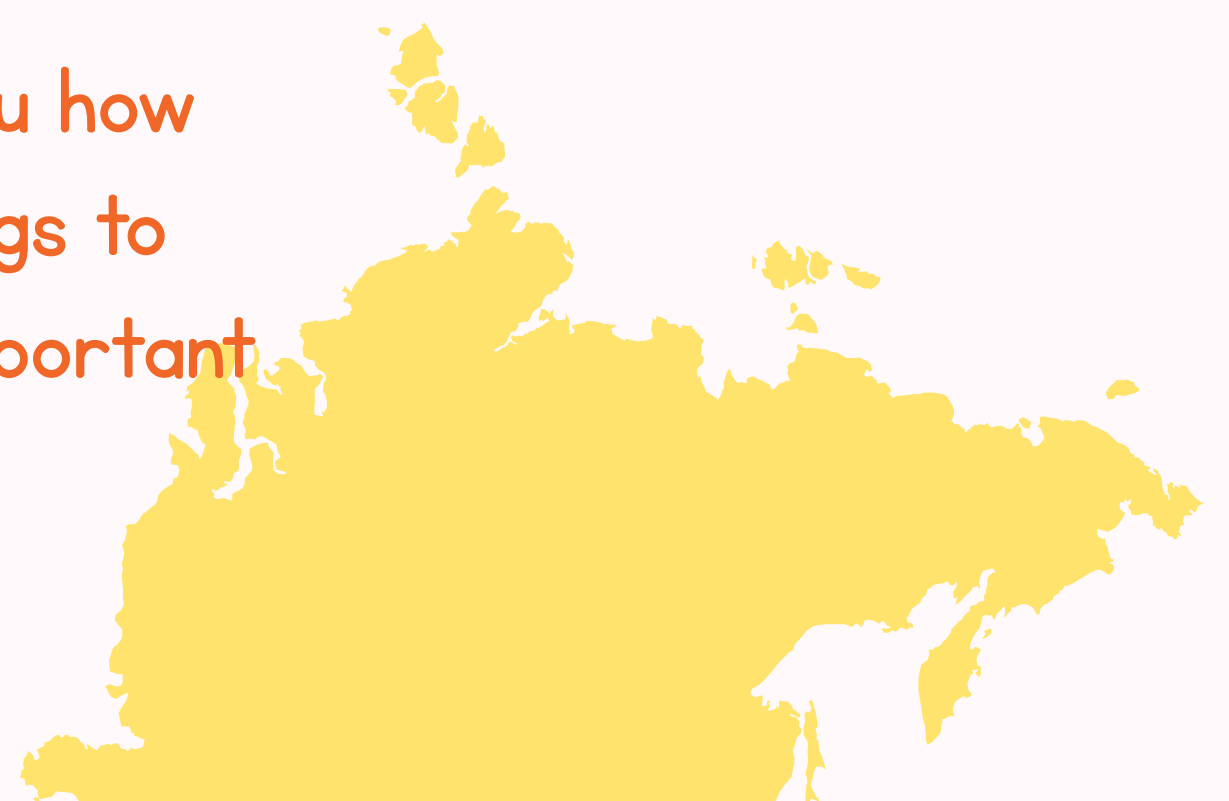
Imagine a world without maps.

We would not know where to go, how to travel, or how our planet looks.

Cartography and mapmaking make this possible.

They allow us to explore, plan, and connect places around the world.

In our presentation, we will show you how maps have changed from old drawings to digital tools, and why they are so important today, especially in Senegal.



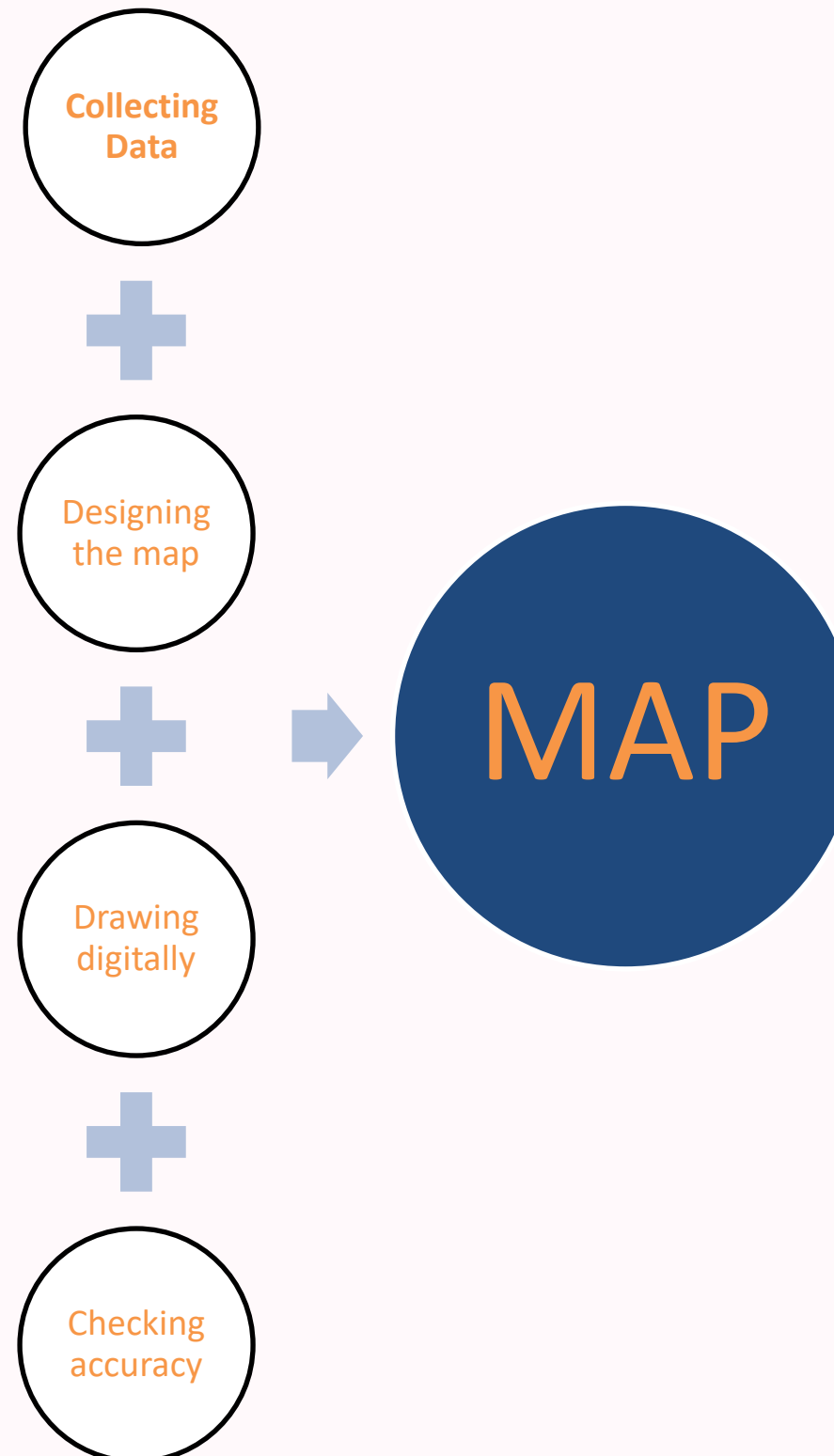
History and Defintion

The history of cartography is very old. The first maps were made thousands of years ago on clay, stones, or animal skins. In ancient Egypt, Greece, and Rome, maps were used for travel and trade. During the Middle Ages, maps were based more on religion than on science. In the Renaissance, with explorers like Columbus, maps became more accurate. Today, modern cartography uses satellites, computers, and GPS to create digital maps. Cartography is the art and science of making maps to represent the Earth and its features.



What is Mapmaking ?

Mapmaking is the process of creating maps using data, symbols, and design to show places on Earth.



Tools, Technologies and Users

Material tools

Used during field data collection

- GPS
- Compass
- Rangefinder or measuring tape
- Drones or satellite images
- Field notebook or tablet



Tools, Technologies and Users

Software tools

Used to process, analyze, and represent data:

- GIS (Geographic Information Systems):

QGIS , ArcGIS , Google Earth Pro, etc.

- Online visualization and mapping tools:

Google My Maps, OpenStreetMap, Mapbox, CartoDB, etc.

- Satellite image processing software:

ENVI, ERDAS Imagine, SNAP (ESA), etc.



Tools, Technologies and Users

USERS

1. Geography and territorial planning
2. Agriculture and environment
3. Risk management and civil protection
4. Urban planning and transportation
5. Tourism and culture
6. Scientific research and education
7. Defense and security

Importance



Advantages and Goals

Clear and synthetic vision

Monitoring and management of natural resources

Represent geographic space

Communicate spatial information

Analyze the space

To find one's bearings at any point in space

Conclusion

In conclusion, cartography and mapmaking are very important in today's world.

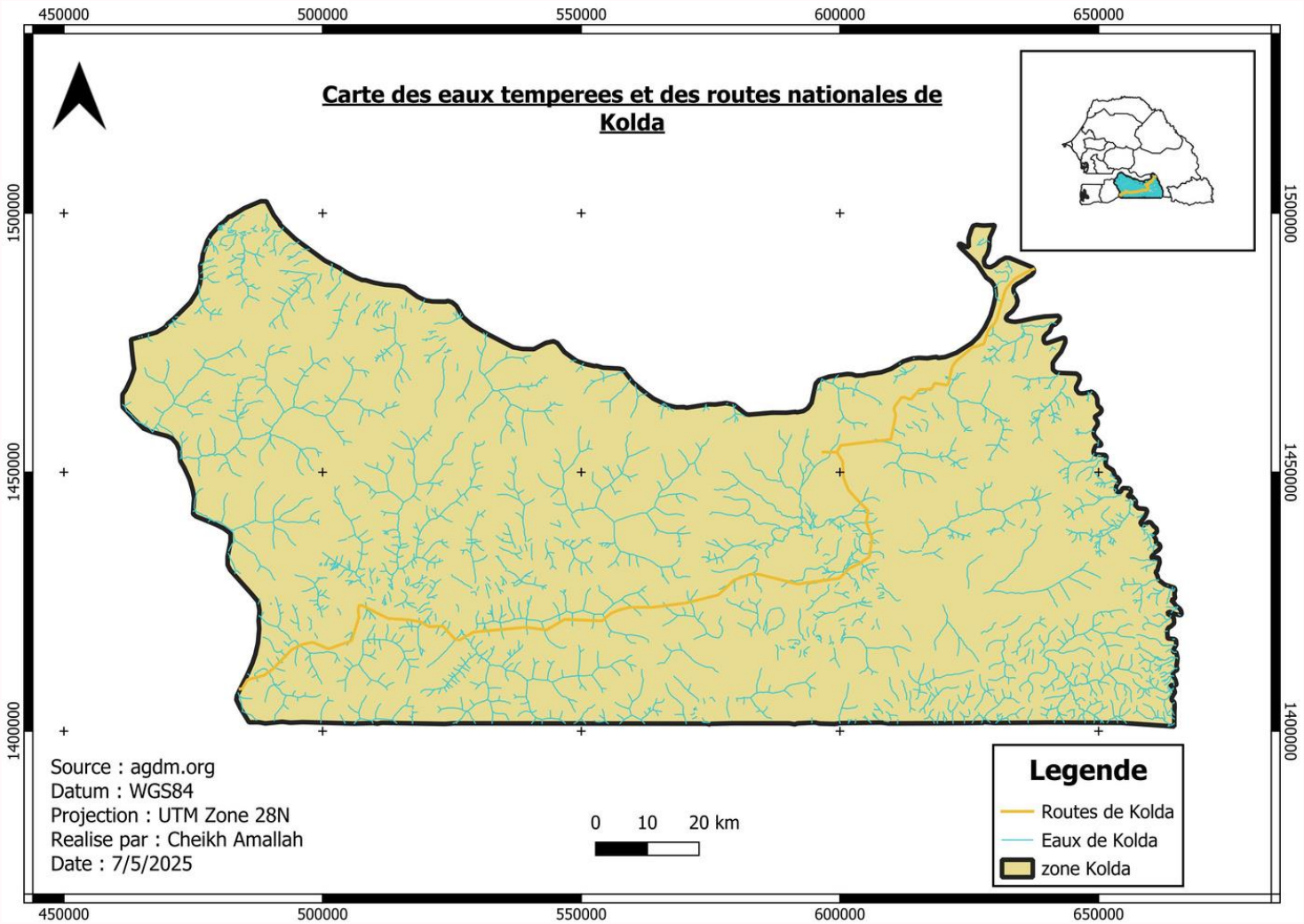
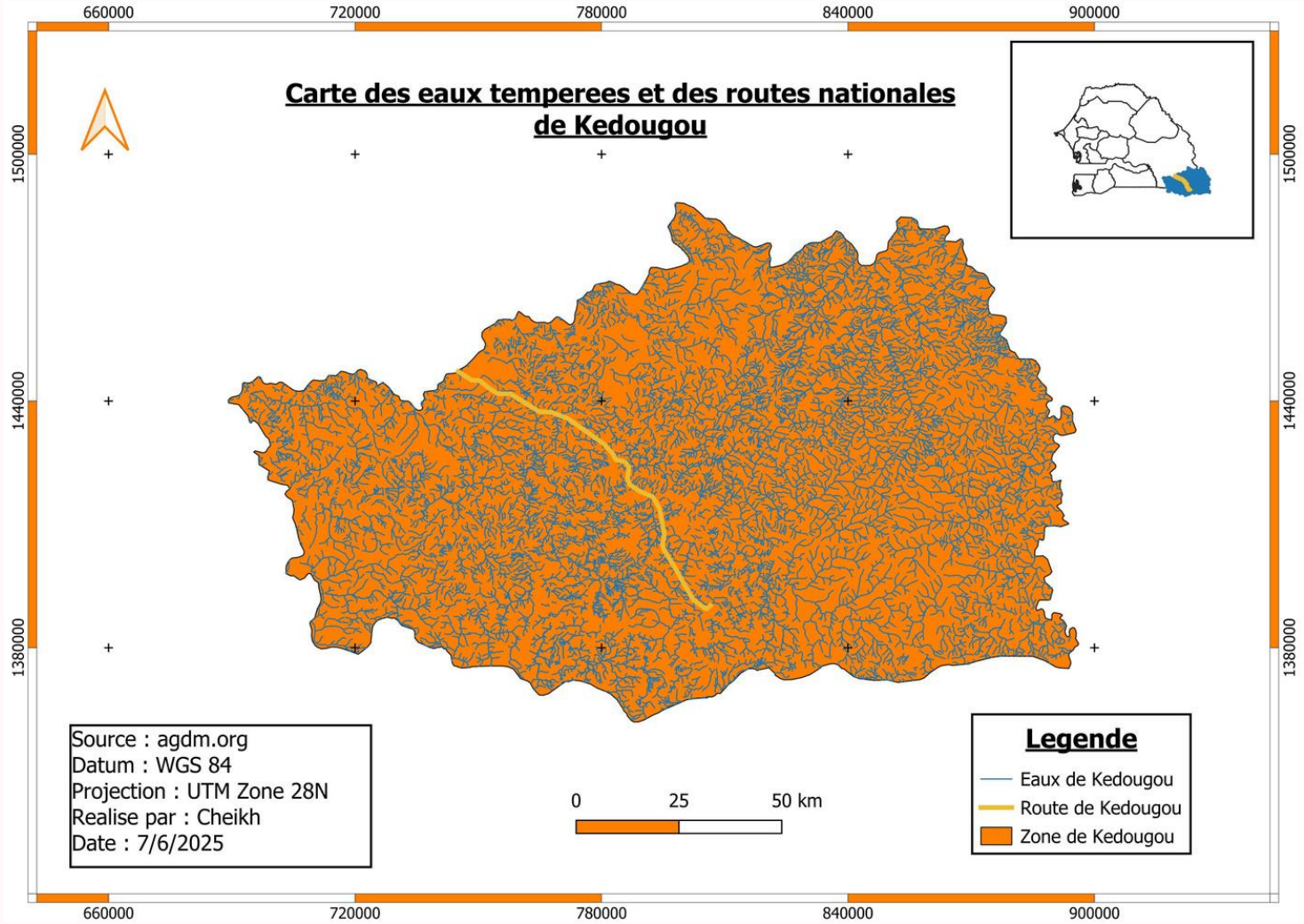
They help us understand our planet, plan our cities, and protect the environment.

In geomatics, cartography is used to collect and analyze data about land, water, and population. It helps governments and organizations make better decisions.

In Senegal, cartography plays a key role in city planning, agriculture, and disaster management for example, to study floods or build new roads. All around the world, maps connect people, improve transport, and support development.

Cartography is not only about drawing maps it is about understanding and shaping our future.

Example of Maps we have made



Kedougou and Kolda

**Thank You for
Attention**

